



PowerShell Conference Europe

Next Up:

Evgenij Smirnov

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PowerShell Conference Europe

Develop, maintain and distribute
PowerShell modules
in isolation

Evgenij Smirnov

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Devolutions



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SynEdgy



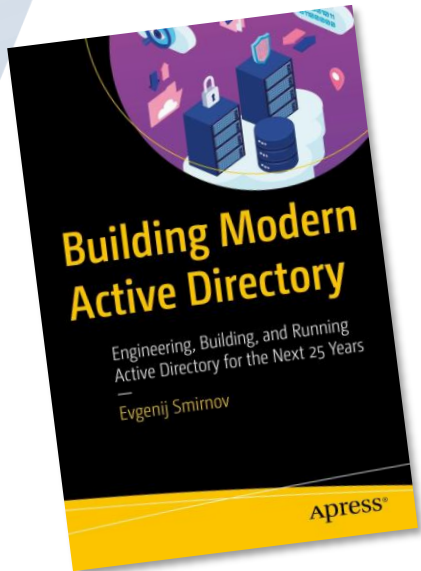
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SPEAKER

Evgenij Smirnov

- I live in Berlin
- Principal Solutions Architect @ Semperis
- 30+ years of consulting and delivery
- PowerShell MVP since 2020
- Attended all PSConfEU since 2016
 - ...and spoke at seven of them, including this one
- AMA about Active Directory
 - ...chances are, you'll not like the answer



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In this session

- Make 'em
- Store 'em
- Update 'em
- Ship 'em!

Not in this session

- Best, or any, practices for module design
- Containers (although they are super helpful!)
- Probabilistic text-extrusion machinery™

Setting the stage



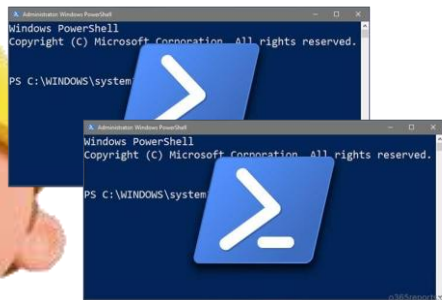
Alice



Craig

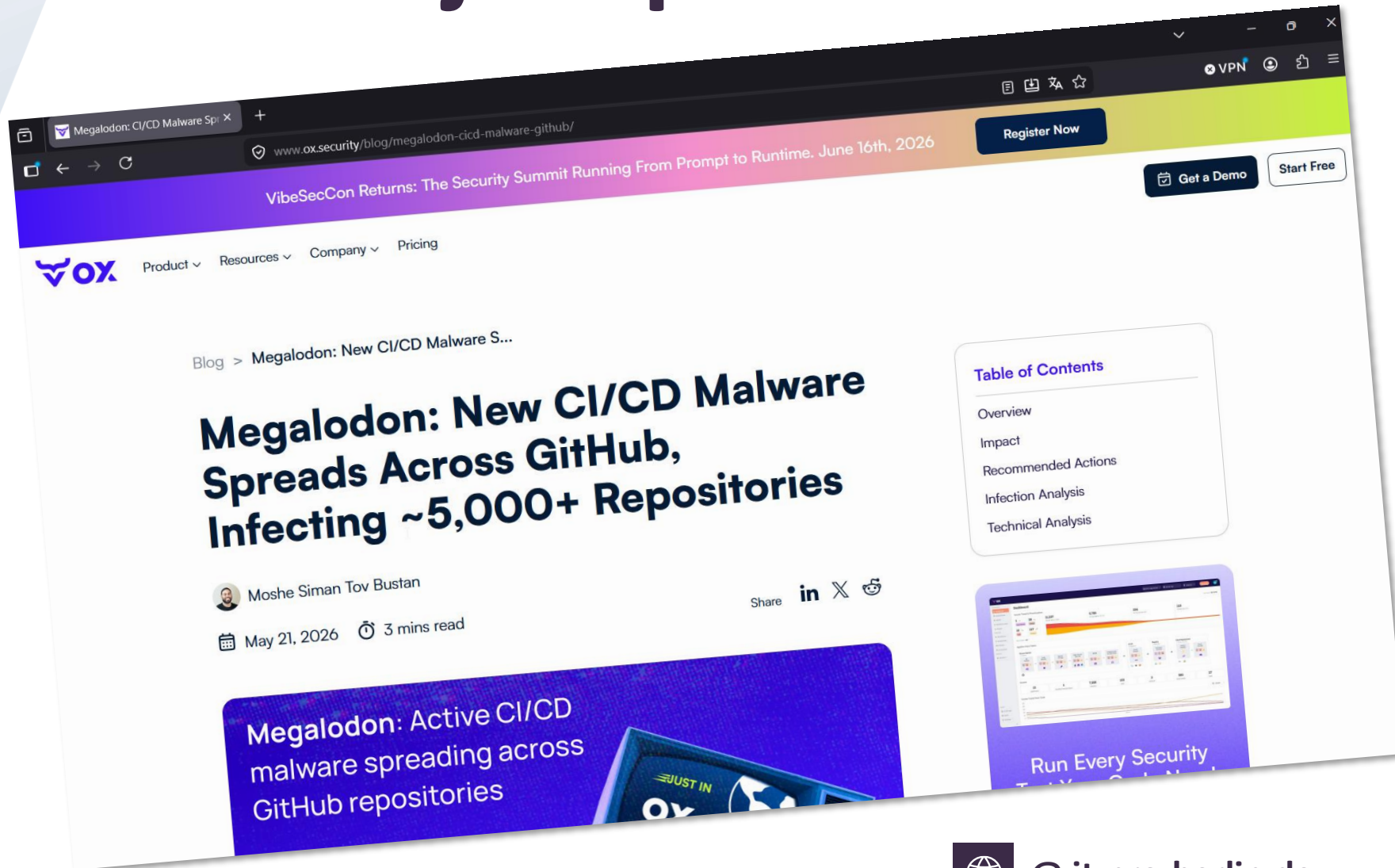


Bob



Dora

...can we just put it on GitHub?



The screenshot shows a web browser displaying a blog post from Ox Security. The browser's address bar shows the URL `www.ox.security/blog/megalodon-cicd-malware-github/`. The page features a navigation bar with the Ox Security logo and links for Product, Resources, Company, and Pricing. A banner at the top of the page reads "VibeSecCon Returns: The Security Summit Running From Prompt to Runtime. June 16th, 2026" with buttons for "Register Now", "Get a Demo", and "Start Free". The main article title is "Megalodon: New CI/CD Malware Spreads Across GitHub, Infecting ~5,000+ Repositories" by Moshe Siman Tov Bustan, dated May 21, 2026, with a 3-minute read time. A "Table of Contents" sidebar lists: Overview, Impact, Recommended Actions, Infection Analysis, and Technical Analysis. At the bottom, there are two featured images: one with the text "Megalodon: Active CI/CD malware spreading across GitHub repositories" and another showing a security dashboard with the text "Run Every Security".

Blog > Megalodon: New CI/CD Malware S...

Megalodon: New CI/CD Malware Spreads Across GitHub, Infecting ~5,000+ Repositories

Moshe Siman Tov Bustan

May 21, 2026 3 mins read

Share in X

Table of Contents

- Overview
- Impact
- Recommended Actions
- Infection Analysis
- Technical Analysis

Megalodon: Active CI/CD malware spreading across GitHub repositories

Run Every Security



Demos

You asked for it...



Chaos → Order

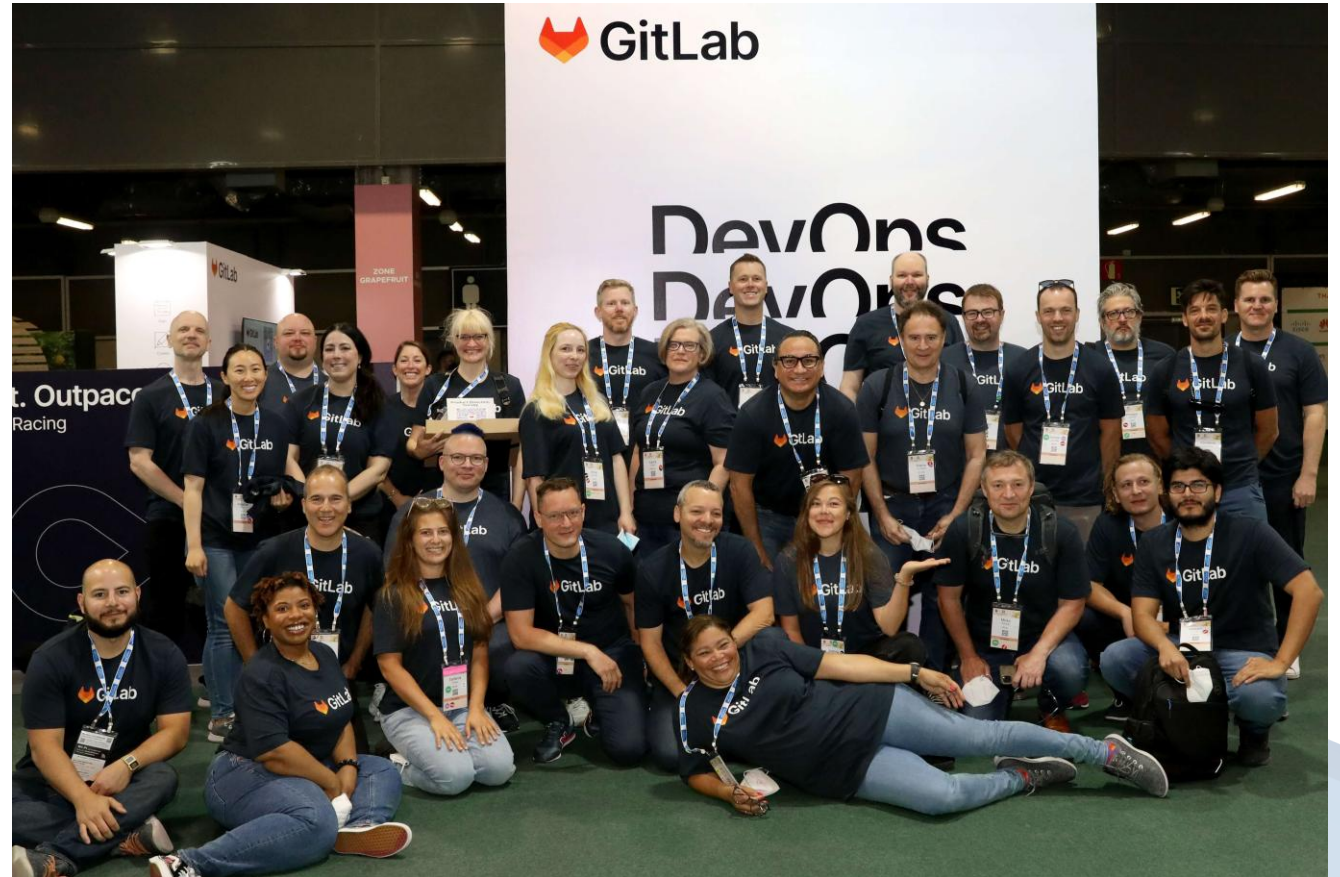
Alice and Bob put their work in source control



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Thank you GitLab!

- Self-hosted
 - if you are so inclined
- Free
 - with limitations
- Super mature
- Doesn't phone home
- Well-Documented



Laying a foundation for PSFactory

- GitLab is a complex toolstack
- The installation is, however, super easy
 - if you start with a supported Linux distribution...
 - ...prepare the prerequisites in advance...
 - ...and don't try to get clever
- I wrote up the setup process in a blog post:
<https://it-pro-berlin.de/2026/06/powershell-module-factory/>



Source control is great!

- We don't need to use `This.Totally.ScoutsHonor.ps1` anymore!
- But what if we could build the modules automatically?

Building & Versioning modules

Alice and Bob dive into CI/CD...



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CI/CD is fun 😊

- This is so cool, the module builds automagically!
- But how do we keep track of versioning?

Keeping track of UseThis.ps1

- Didn't Emrys suggest something, like, two years ago?
- Sure! But let's take a slightly different approach
 - We will version EVERY build in the Patch number...
 - ...and reset on major and minor changes!
- OK, but this only works if Alice and Bob do not work in parallel on the same project!
 - There is, of course, a way more elegant DevOps'y approach to this 😊

CI/CD variables are cool!

- Let's get rid of moduleinfo.json for pipeline builds!
- But we still need it for local builds!

Creating artifacts

Because %TEMP% is not very human-friendly



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Artifacts for humans

- Now we know where to find the built modules!
- But how can we ensure no one changes them after we give them to Craig and Dora?

Signing the modules

Because... integrity?



Publishing the modules

So that Craig and Dora can find them



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Publishing in the project

- \$CI_JOB_TOKEN is a valid authenticator here!
- But we'll end up with 20 repos for 20 modules...

...not cool.



Central publishing

- The central registry MUST have limited visibility!
 - ...because only then anonymous pull is available

What about third-party stuff?

- Well, a NuGet package is a NuGet package 😊

Deploying the consumer config

So that Craig and Dora don't need to worry



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What works and what doesn't?

- With PSResourceGet and default v3 feed: everything works
- With PowerShellGet and (new) v2 feed:
 - Publishing works
 - Retrieval does not (no search capabilities in endpoint)

Scaffolding the whole thing

Because it was a lot of clicks...



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Module scaffolding in PSFactory

- Create a project
 - Create a project in GitLab
 - Create a local git folder
 - Create the build script, the parts folder and CI/CD config
- Create project variables
 - Module name, GUID, description, versioning info
- Create access token for versioning ← this stopped working 😞
 - ...and store it in a variable



Q&A

15 minutes



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GIVE ME FEEDBACK



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isolation



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